
What is good open source?
Richard Littauer - http://richard.social
Rimu https://www.inaturalist.org/observations/198921399

Who is this Richard Littauer, anyway

- Linguist and computer scientist
- Software developer
- Community manager
- Worked for: Hipcamp, MIT, Johns Hopkins, Protocol Labs, Open Source Initiative (OSI), Open Source Collective, the Digital Infrastructure Fund, GNOME…
- Currently:
  - **CURIOSS**
  - **SustainOSS**
  - **Language consultant**

Source not given

Also a PhD student



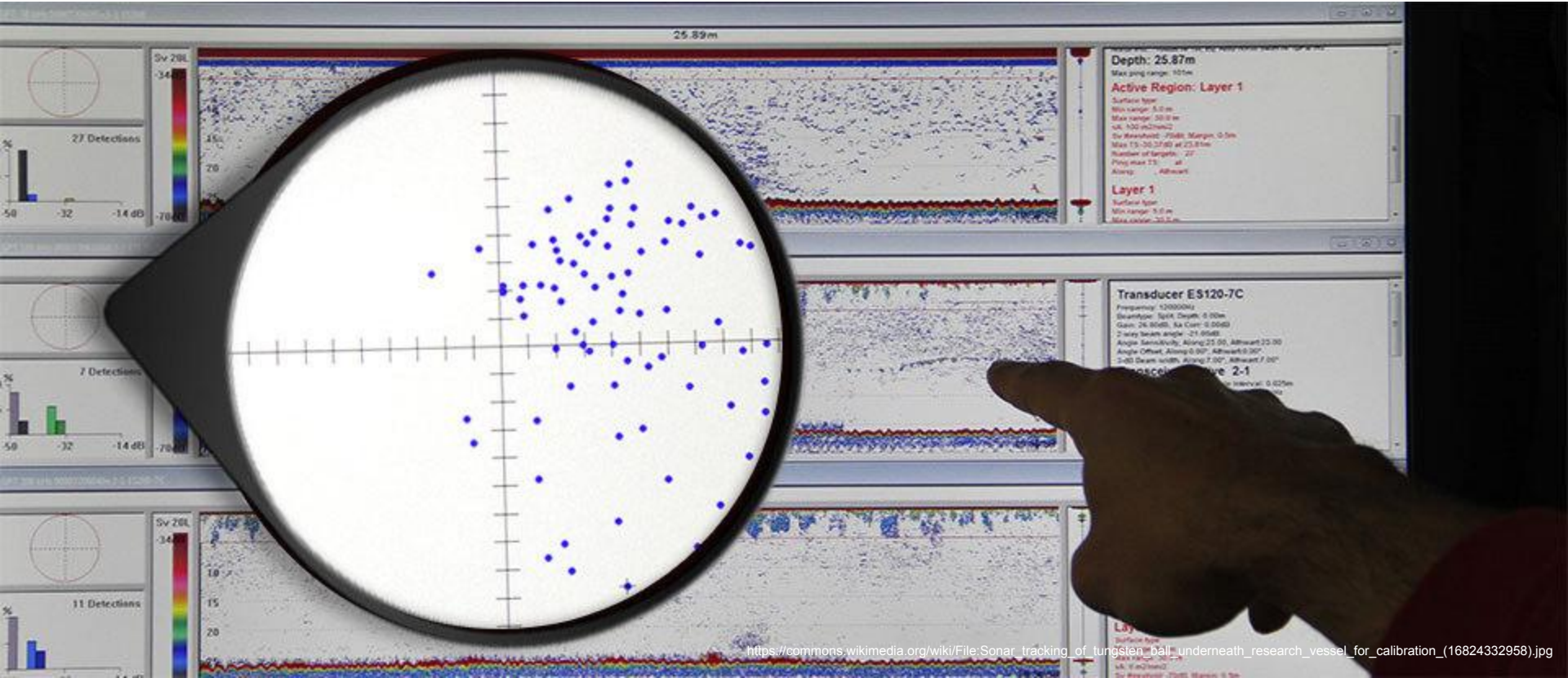
Silver Gull https://burntfencreativellc.pixieset.com/newzealand/

What is Open Source

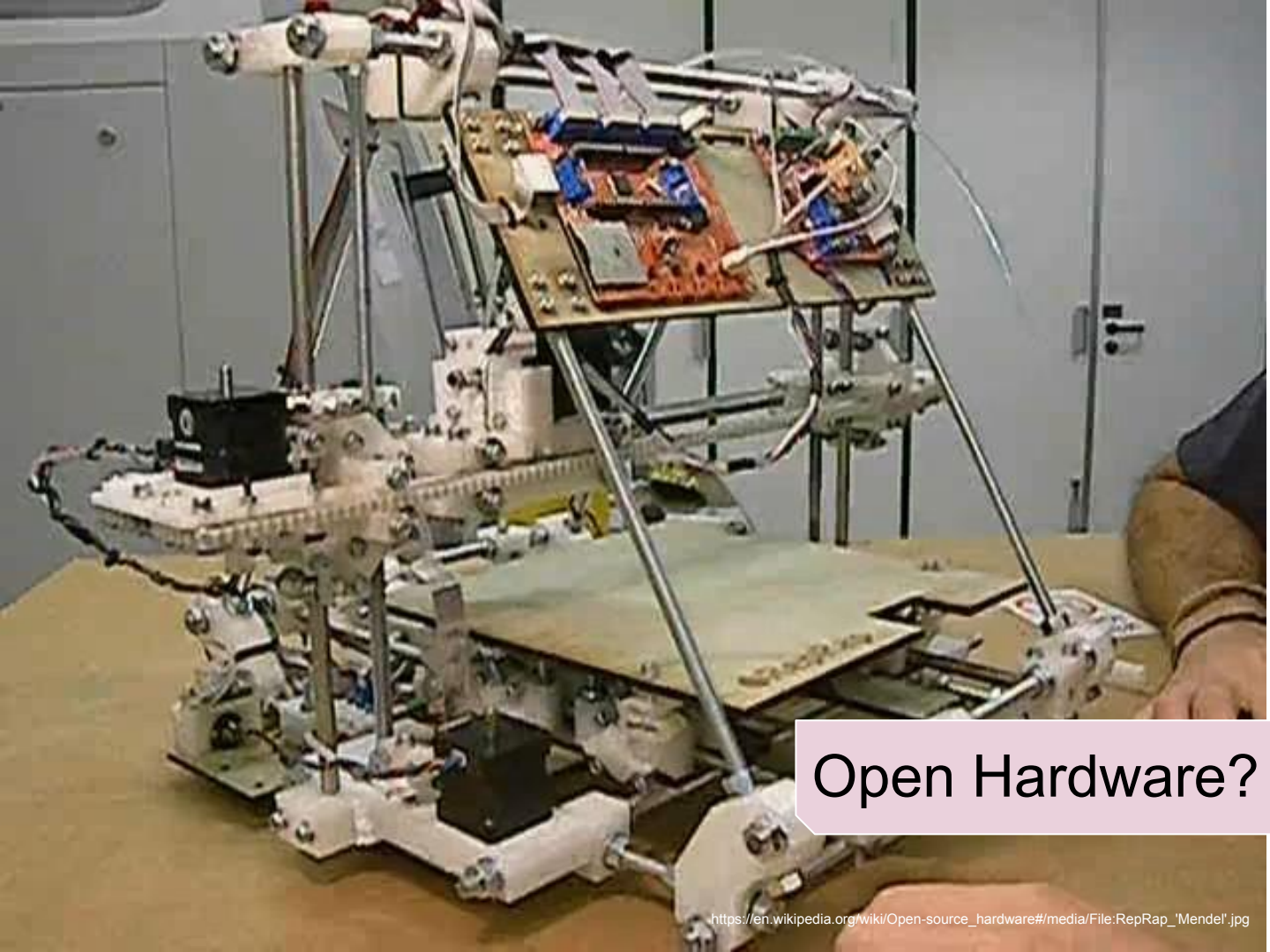


Paradise Shelduck https://www.inaturalist.org/observations/54206748

Open Data?

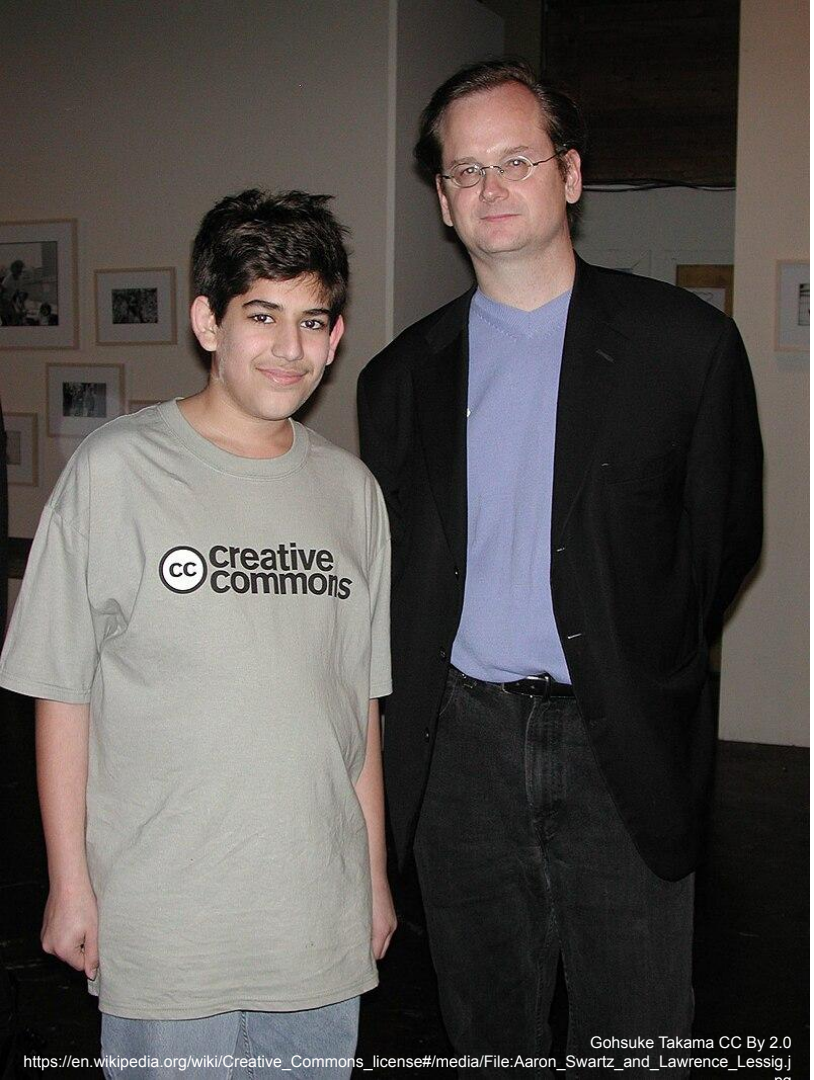


https://commons.wikimedia.org/wiki/File:Sonar_tracking_of_tungsten_ball_underneath_research_vessel_for_calibration_(16824332958).jpg



Open Hardware?

https://en.wikipedia.org/wiki/Open-source_hardware#/media/File:RepRap_'Mendel'.jpg



Gohsuke Takama CC By 2.0
https://en.wikipedia.org/wiki/Creative_Commons_license#/media/File:Aaron_Swartz_and_Lawrence_Lessig.jpg

Creative Commons?

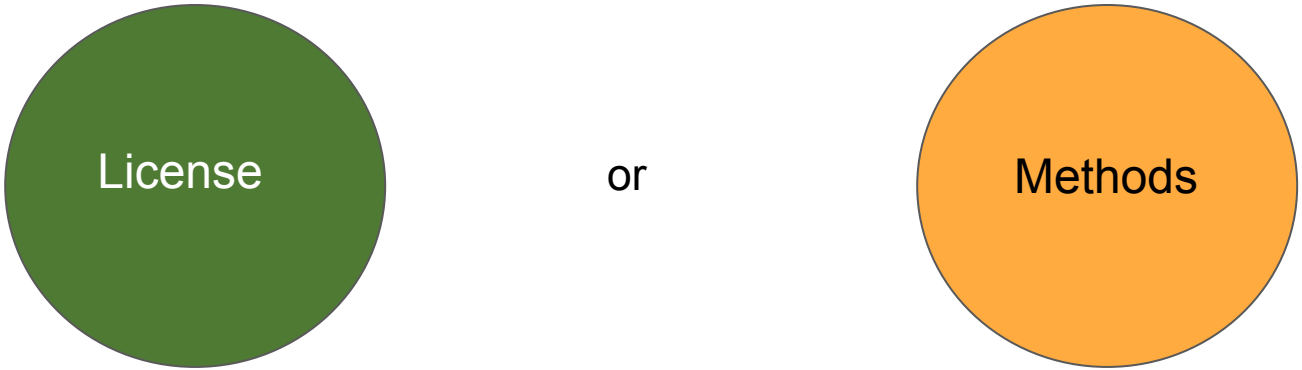
Other opens

- Open Science
- Open Scholarship
- Open Education
- Open Government
- Open Strategy
- Open Innovation
- Open Platform
- Open Access
- Open Organizing
- …

What we're going to cover

- Basic definition of open source
- What makes OSS healthy or sustainable

Open source:



MIT License

MIT License

Copyright (c) The dog I wish I had 2025

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

GPL

- "Copyleft" licenses
- Too long to fit on this slide
- Use it as much as you want, just publish your stuff as open, too.

Open source:



open source initiative®

Methods

What about?



What about?


Open Source*
Organization for
Ethical Source
Source: Midjourney

How much open source?

10.7 million Packages

230 million Repositories

19 billion Dependencies

1.6 million Developers



How much open source?

GitHub

150+ million	4+ million	420+ million	90%
Developers	Organizations	Repositories	Fortune 100

2024 WORKING PAPER HBS WORKING PAPER SERIES

The Value of Open Source Software

By: Manuel Hoffmann, Frank Nagle and Yanuo Zhou

Format: Print | **Language:** English | **Pages:** 42

Email Print Share

ABSTRACT

The value of a non-pecuniary (free) product is inherently difficult to assess. A pervasive example is open source software (OSS), a global public good that plays a vital role in the economy and is foundational for most technology we use today. However, it is difficult to measure the value of OSS due to its non-pecuniary nature and lack of centralized usage tracking. Therefore, OSS remains largely unaccounted for in economic measures. Although prior studies have estimated the supply-side costs to recreate this software, a lack of data has hampered estimating the much larger demand-side (usage) value created by OSS. Therefore, ge unique global da Ve first estimate th We then calcula software and wo f widely-used OSS nd that firms would The top six program Further, 96% of the demand-side value is created by only 5% of OSS developers.

We estimate the supply-side value of widely-used OSS is $4.15 billion, but that the demand-side value is much larger at $8.8 trillion.

ABOUT THE AUTHOR



Frank Nagle
Strategy
→ More Publications

MORE FROM THE AUTHORS

JANUARY 2025 FACULTY RESEARCH

AGENTS.INC: PATHWAYS TO GROWTH AT AN AI STARTUP

By: Frank Nagle and Susan Pinckney

NOVEMBER 2024 FACULTY RESEARCH

FEDEX CYBERATTACK (B): REFLECTIONS AND LESSONS

By: Hise Gibson, Frank Nagle, Alicia Dadlani and Martha Hostetter

NOVEMBER 2024 FACULTY RESEARCH

FEDEX CYBERATTACK (A): NAVIGATING THE NOTPETYA STORM

By: Hise Gibson, Frank Nagle, Alicia Dadlani and Martha Hostetter

Why would you use Open Source?

Why would you use Open Source?

- You already are.

Why would you produce Open Source?

Why would you produce Open Source?

- Collaborate with others
- More eyes = less bugs
- Improve your CV
- Status and clout
- More adoption
- "Feels good"
- Security (never roll your own)

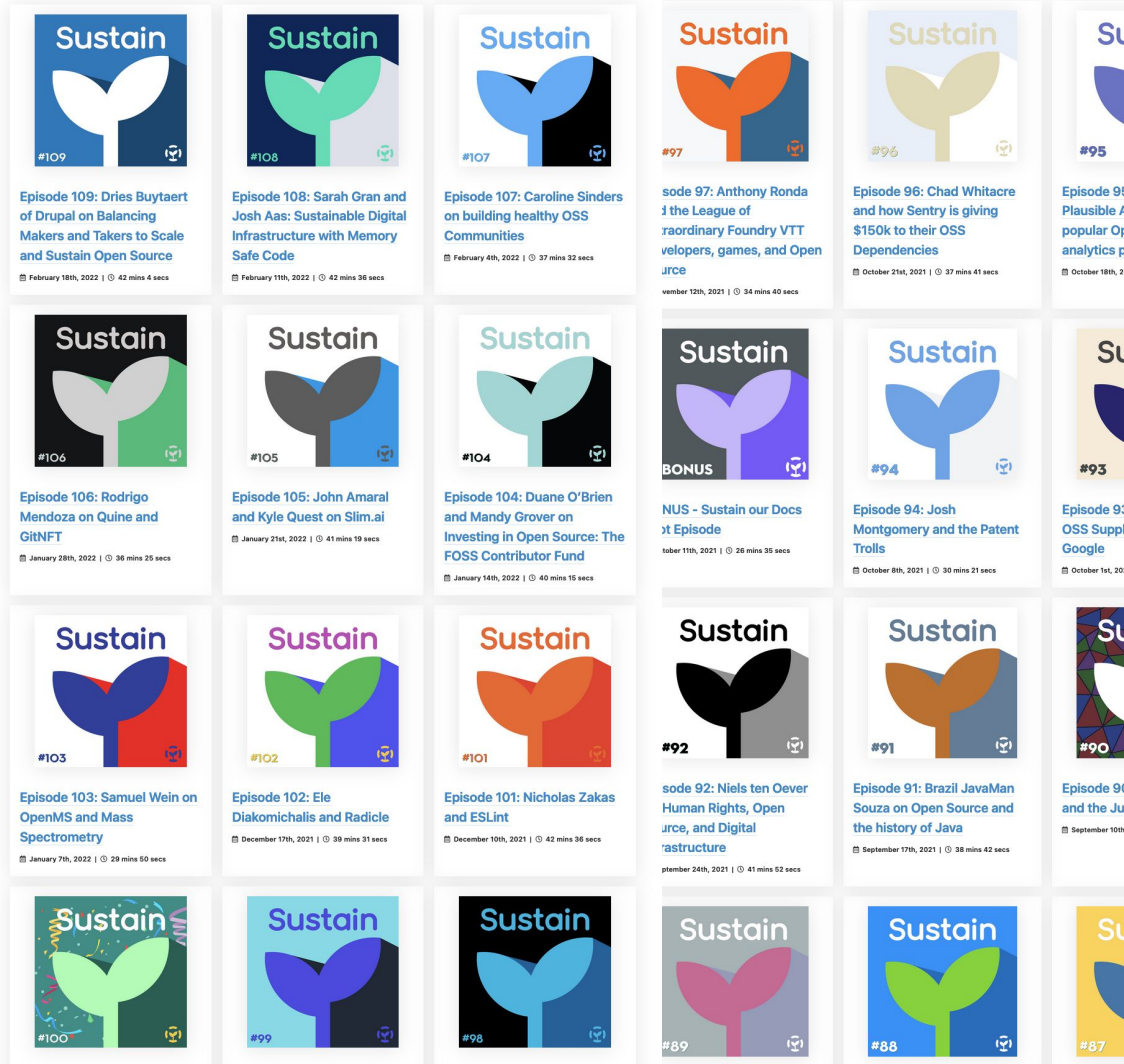
Why would you **NOT** use Open Source?

Why would you **NOT** use Open Source?

- Extractive use cases
- Poor expectation setting for users
- Too transparent (medical, indigenous rights)
- Too much work building community
- Business logic could be proprietary
- Security (never roll your own)

SustainOSS Podcast

- Over 300 podcasts
- Hundreds of guests
- Mostly about open source



thesis Public

Pin | Unwatch 5 | Fork 4 | Starred 20

master | 2 Branches | 0 Tags

RichardLitt Add slides — 75b2b9e · 7 years ago — 233 Commits

img	Finished edits	7 years ago
meeting-notes	chore: update to check string	7 years ago
proposal	Moving proposal to its own folder	7 years ago
reading-notes	Change apostrophes	7 years ago
.gitignore	Use a proper thesis style. Thanks @jonathanPoitz	8 years ago
.travis.yml	fix: links	7 years ago
LREC2016.zip	Archived to stop text issues	7 years ago
README.md	docs: Add note about where to find the file	7 years ago
abstract.tex	Edit document after Alexis's comments	7 years ago
acknowledgements.tex	Small fixes	7 years ago
case-studies.tex	Edits to the Naskapi section	7 years ago
colloquium_slides.key	Add slides	7 years ago
colloquium_slides.pdf	Add slides	7 years ago
commands.sh	Add all commands into a shell file	7 years
conclusion.tex	Finished edits	7 years
data.json	Add all commands into a shell file	7 years
declaration.tex	Use roman characters for initial pages	7 years ago

My thesis on "Open Source Code and Low Resource Languages" for an MSc in Language Science and Technology at Saarland University

nlp thesis dissertation saarland endangered-languages nlproc lrl saarland-university low-resource-languages

Readme
Activity
20 stars
5 watching
4 forks

Contributors 2

RichardLitt Richard Littauer
fritzvd Fritz van Deventer

TeX 99.5% Shell 0.5%

Does sustainability matter for this repo?

perisoreus Public

Pin | Unwatch 1 | Fork 1 | Star 2

main | 1 Branch | 1 Tag

RichardLitt Tighten up figure — b409718 · 5 months ago — 13 Commits

.gitignore	Adding sound files	6 months ago
558679851.wav	Adding sound files	6 months ago
585940541.m4a	Adding sound files	6 months ago
585940541.wav	Adding sound files	6 months ago
Figure.png	Tighten up figure	5 months ago
LICENSE	Init	7 months ago
README.md	Init	7 months ago
appendix-1.md	Add in appendix	6 months ago
create_figures.py	Tighten up figure	5 months ago
get_attribution_for_sic_jay.py	Init	7 months ago
get_data_from_macaulay_library.py	Init	7 months ago
get_gryjay_species_comments.py	Init	7 months ago

Scripts used on files from various community science platforms, to study vocalisations in Perisoreus species

nlp ebird inaturalist community-science xeno-canto

Readme
MIT license
Activity
2 stars
1 watching
1 fork

1 tags

Contributors 2

RichardLitt Richard Littauer
dbetchkal Davyd Halyn Betchkal

Languages

Does sustainability matter for this repo?

curl Public

Sponsor | Watch 783 | Fork 6.7k | Star 37.7k

master | 22 Branches | 223 Tags

bagder docs/libcurl: fix type and prototype problems in examples — b1eebdf · 10 minutes ago — 35,021 Commits

.circleci	GHA: update wolfSSL/wolfssl to v5.8.0	4 days ago
.github	docs/libcurl: fix type and prototype problems in examples	10 minutes ago
CMake	build: enable gcc-15 picky warnings	3 hours ago
LICENSES	copyright: update all copyright lines and remove year ran...	3 years ago
docs	docs/libcurl: fix type and prototype problems in examples	10 minutes ago
include	typecheck-gcc.h: fix the typechecks	5 days ago
lib	scripts: fix perl indentation, whitespace, semicolons	3 hours ago
m4	build: enable gcc-15 picky warnings	3 hours ago
packages	websocket: add option to disable auto-pong reply	2 weeks ago
plan9	copyright: update all copyright lines and remove year ran...	3 years ago
projects	curltime: use libcurl time functions in src and tests/server	last month
scripts	scripts: fix perl indentation, whitespace, semicolons	3 hours ago
src	lib: provide a getaddrinfo wrapper	yesterday
t	...ns	3 hours ago
w	...OME	3 weeks ago
..	...ear ran...	3 years ago
.git-blame-ignore-revs	copyright: update all copyright lines and remove year ran...	3 years ago

Does sustainability matter for this repo?

A command line tool and library for transferring data with URL syntax, supporting DICT, FILE, FTP, FTPS, GOPHER, GOPHERS, HTTP, HTTPS, IMAP, IMAPS, LDAP, LDAPS, MQTT, POP3, POP3S, RTMP, RTMPS, RTSP, SCP, SFTP, SMB, SMBS, SMTP, SMTPS, TELNET, TFTP, WS and WSS. libcurl offers a myriad of powerful features

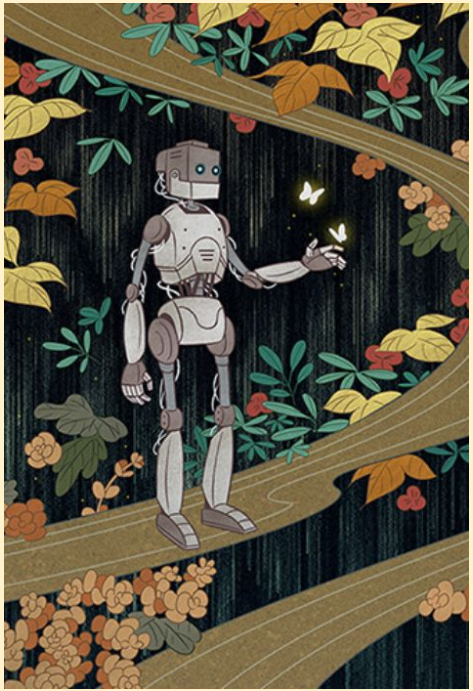
curl.se/

c | http | mqtt | client | ldap | library | user-agent | curl | https | websocket | ftp | sftp | libcurl | scp | pop3 | gopher | transfer-data | hacktoberfest | imaps | transferring-data

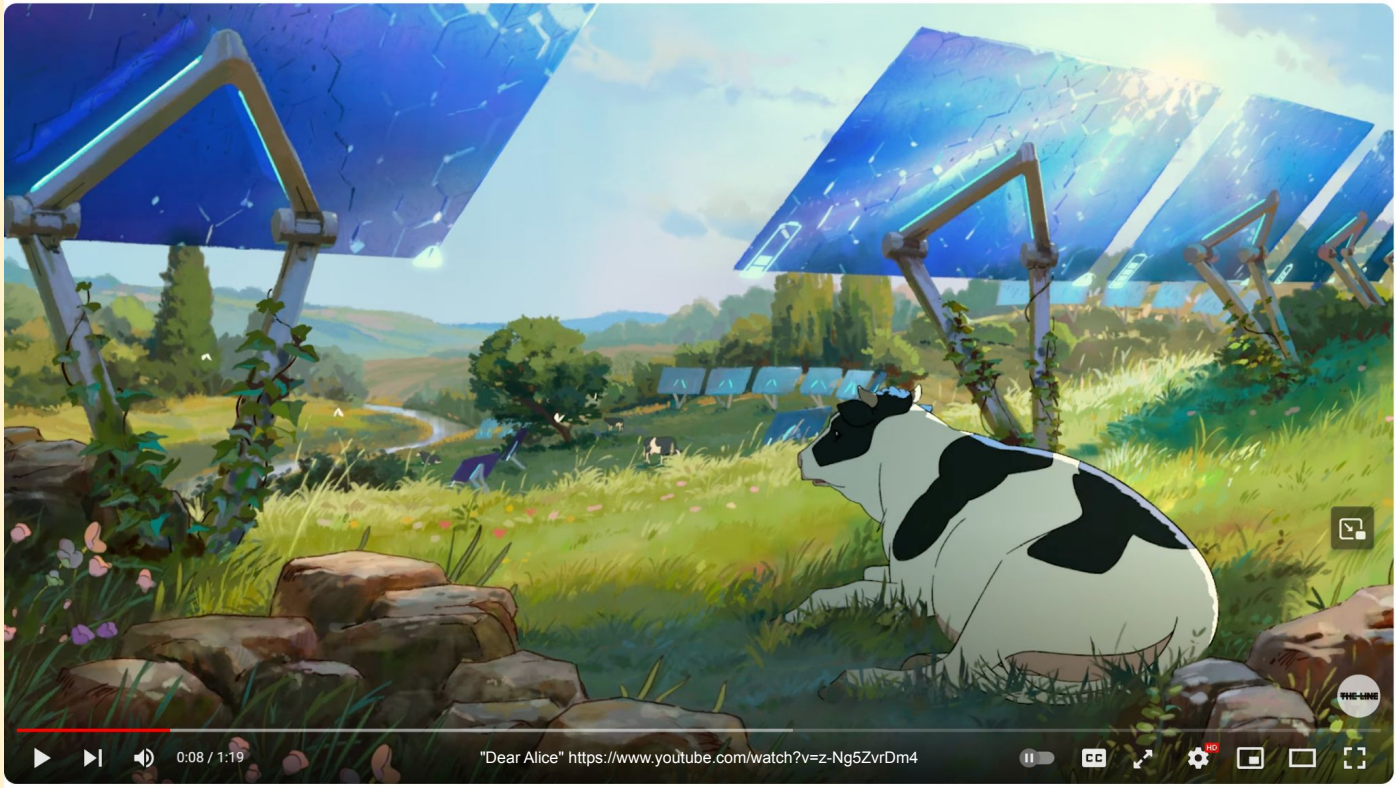
Readme
View license
Code of conduct
Security policy
Activity
Custom properties
37.7k stars
783 watching
6.7k forks

8.13.0 Latest
28 days ago

What is Sustainability?



https://www.behance.net/gallery/127556565/Book-Cover-A-MONK-AND-ROBOT-BOOK



Where sustainability applies in FOSS:

- Projects
- Developers
- Communities
- Businesses
- Academia
- Ecosystems
- Environment



Sustain: FOSS Projects

- Growth
- Diversity
- Funding

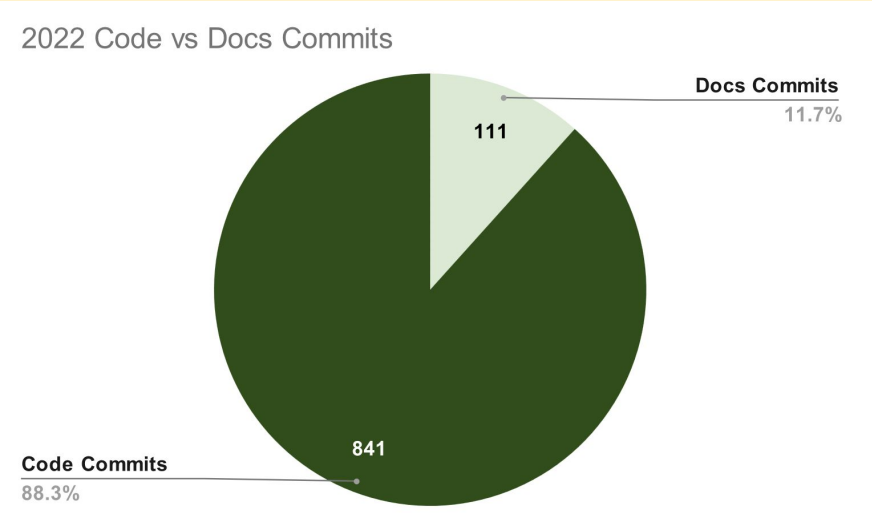
Sustain: FOSS Projects

- Growth
  - More in than out
  - Pathways to maintainership
  - Product management and design planning
  - Goal: capturing market share of attention



Sustain: FOSS Projects

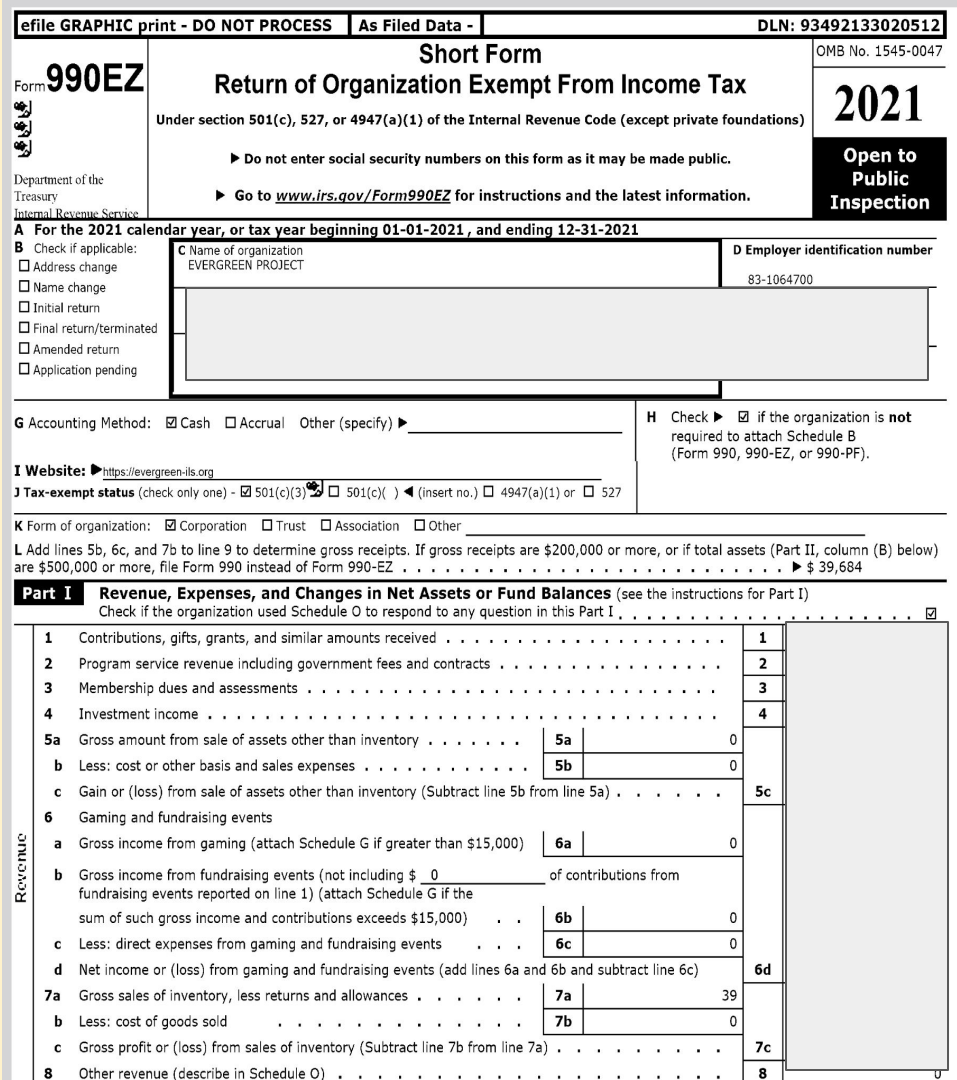
- Diversity
  - Welcoming community spaces
  - CoCs
  - Documentation
  - Moderation
  - Mentoring
  - Goal: towards some ethical destination



https://evergreen-ils.org/wp-content/uploads/2023/12/Annual%20Report%202022.pdf

Sustain: FOSS Projects

- Funding
  - Funding for the project costs
  - Funding for the core maintainers
  - Lifestyle funding
  - Funding for all contributors
  - Non-fiscal sponsorships



efile GRAPHIC print - DO NOT PROCESS | As Filed Data - | DLN: 93492133020512

Form **990EZ**

Department of the Treasury
Internal Revenue Service

Short Form
Return of Organization Exempt From Income Tax

Under section 501(c), 527, or 4947(a)(1) of the Internal Revenue Code (except private foundations)

▶ Do not enter social security numbers on this form as it may be made public.

▶ Go to www.irs.gov/Form990EZ for instructions and the latest information.

OMB No. 1545-0047

2021

Open to Public Inspection

A For the 2021 calendar year, or tax year beginning 01-01-2021 , and ending 12-31-2021

B Check if applicable:
☐ Address change
☐ Name change
☐ Initial return
☐ Final return/terminated
☐ Amended return
☐ Application pending

C Name of organization
EVERGREEN PROJECT

D Employer identification number
83-1064700

G Accounting Method: ☑ Cash ☐ Accrual Other (specify) ▶

H Check ▶ ☑ if the organization is **not** required to attach Schedule B (Form 990, 990-EZ, or 990-PF).

I Website: ▶https://evergreen-ils.org

J Tax-exempt status (check only one) - ☑ 501(c)(3) ☐ 501(c)( ) ◀ (insert no.) ☐ 4947(a)(1) or ☐ 527

K Form of organization: ☑ Corporation ☐ Trust ☐ Association ☐ Other

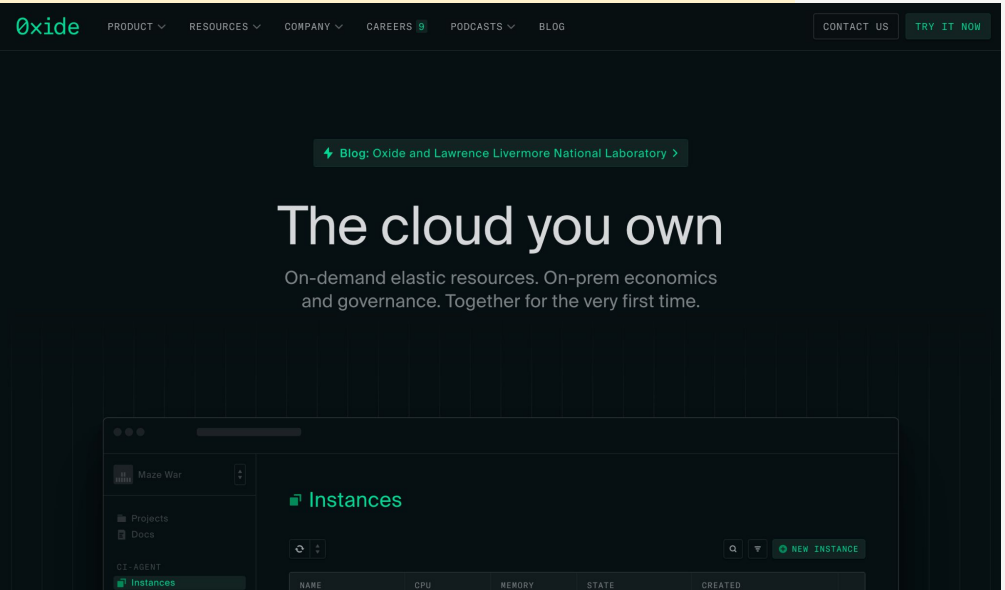
L Add lines 5b, 6c, and 7b to line 9 to determine gross receipts. If gross receipts are $200,000 or more, or if total assets (Part II, column (B) below) are $500,000 or more, file Form 990 instead of Form 990-EZ . . . . . ▶ $ 39,684

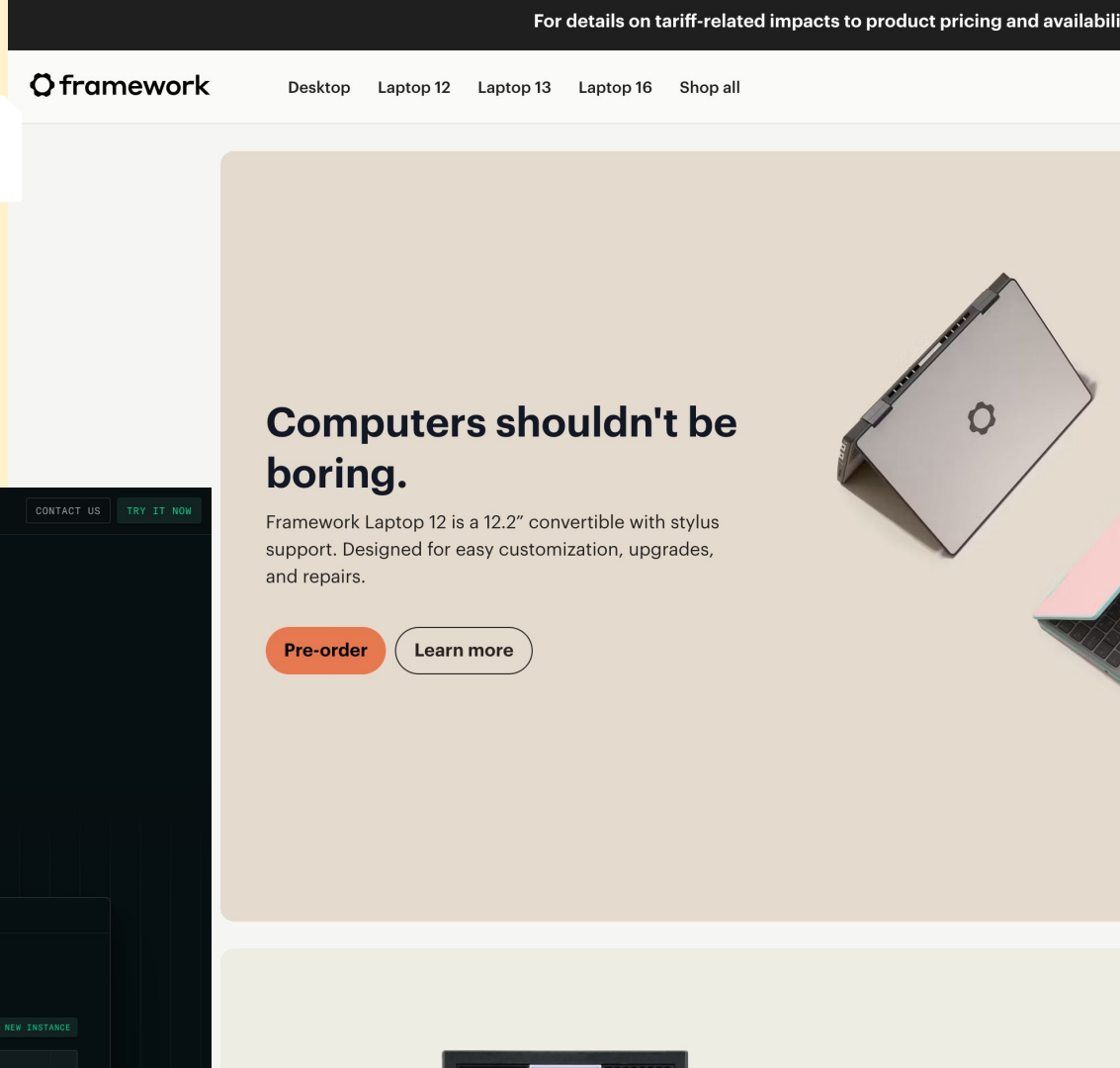
Part I **Revenue, Expenses, and Changes in Net Assets or Fund Balances** (see the instructions for Part I)
Check if the organization used Schedule O to respond to any question in this Part I . . . . . ☑

Revenue	1	Contributions, gifts, grants, and similar amounts received			1	
	2	Program service revenue including government fees and contracts			2	
	3	Membership dues and assessments			3	
	4	Investment income			4	
	5a	Gross amount from sale of assets other than inventory	5a	0		
	b	Less: cost or other basis and sales expenses	5b	0		
	c	Gain or (loss) from sale of assets other than inventory (Subtract line 5b from line 5a)			5c	
	6	Gaming and fundraising events				
	a	Gross income from gaming (attach Schedule G if greater than $15,000)	6a	0		
	b	Gross income from fundraising events (not including $ 0 of contributions from fundraising events reported on line 1) (attach Schedule G if the sum of such gross income and contributions exceeds $15,000)	6b	0		
	c	Less: direct expenses from gaming and fundraising events	6c	0		
	d	Net income or (loss) from gaming and fundraising events (add lines 6a and 6b and subtract line 6c)			6d	
	7a	Gross sales of inventory, less returns and allowances	7a	39		
	b	Less: cost of goods sold	7b	0		
	c	Gross profit or (loss) from sales of inventory (Subtract line 7b from line 7a)			7c	
	8	Other revenue (describe in Schedule O)			8	0

Sustain: FOSS Projects

- You can sell stuff:





Sustain: FOSS Developers

- Satisfaction
- Career trajectories
- Mentoring
- Funds and recognition
- Avoiding burnout



Sustain: FOSS Communities

- Not always tied to a project
- Functionally similar to what we talked about above
- But at scale.



Sustain: FOSS Businesses and Corporate

- Focus: a bottom line
- Not pushing OSS onto your employee's private time
- Having advocates and marketing inside the company for open source
- Risk
- OSPOs
- "Collaboration"

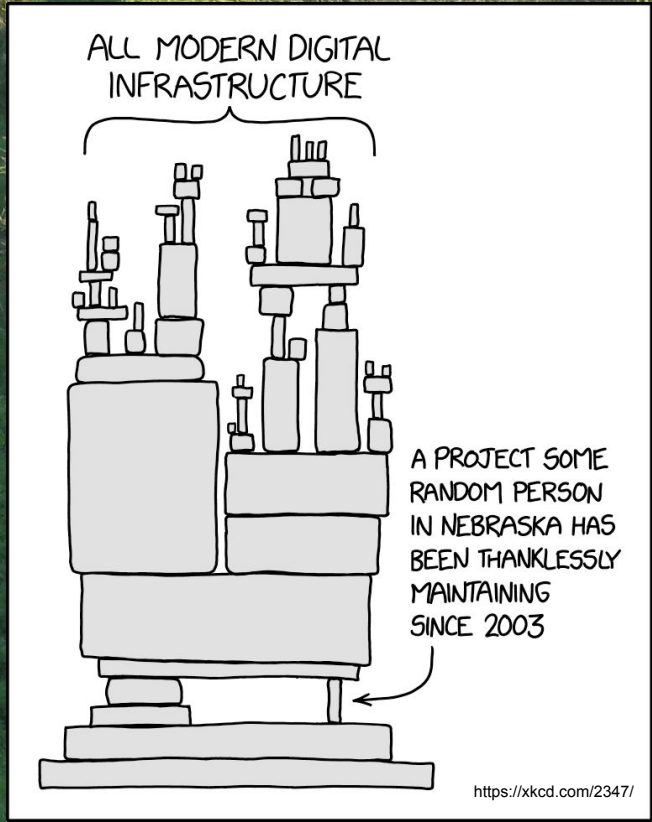


Sustain: FOSS Academia

- Not quite the same as industry or most open source
- Reproducibility
- Citations and altmetrics
- The project- or funding-based cycles of OSS in academia
- The student-graduation cycle of academia
- Where support comes from in the institution

Sustain: FOSS Ecosystem

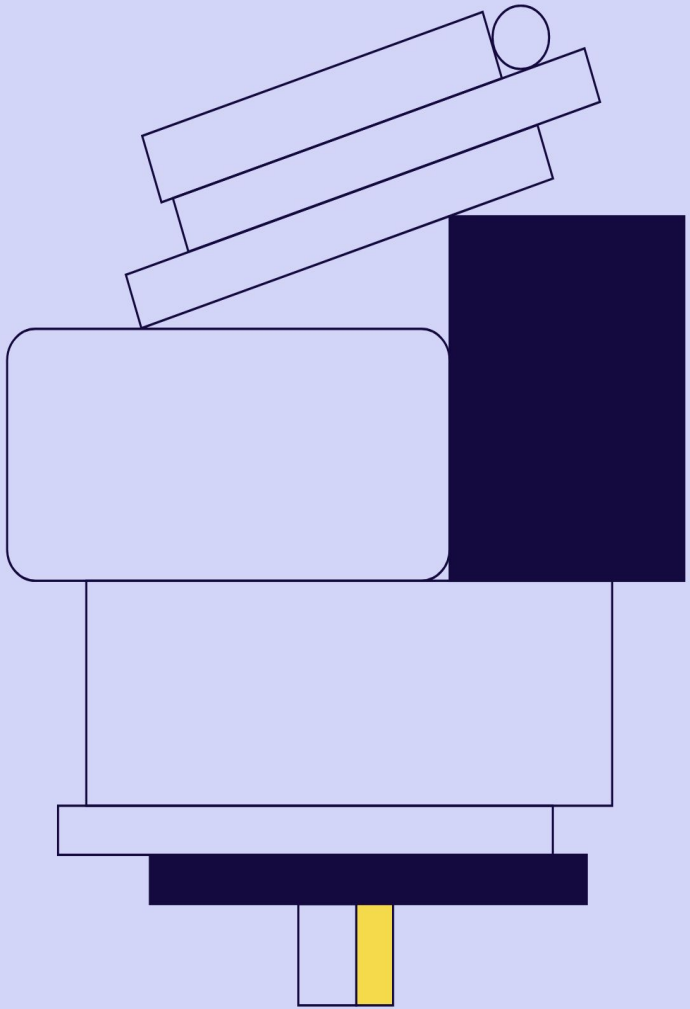
- Funding maintainers for their work
- Security and supply chains
- The "digital commons"



Gorse https://www.inaturalist.org/observations/65500592

Sustain: FOSS Ecosystem

- Legal sustainability and licenses
  - Stopping proliferation
  - Open washing
  - Patent issues
- AI + open source
- Taxation and infrastructural funding
- The CRA and their ilk



https://www.sovereigntechfund.de/

Sustain: FOSS Ecosystem



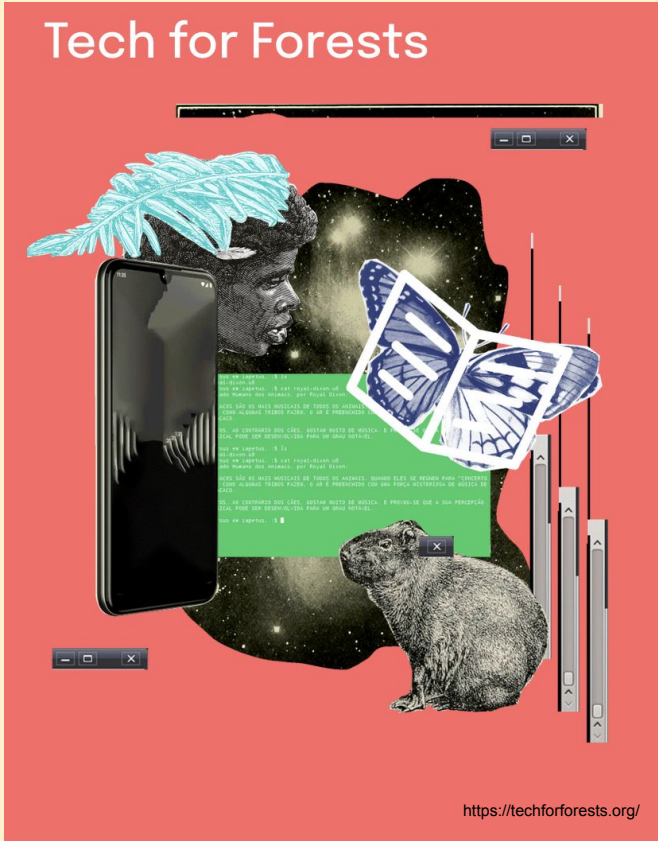
Sustain: FOSS & Environment

- Open source hardware
- Carbon-neutral data storage centers
- Reducing server costs
- Not using bitcoin
- Avoiding the conference trap



Sustain: FOSS & the whole environment

- Tech for environmental activists
- Tech for environmental studies
- Use of tech for the planet as a whole



Sustain: FOSS & the rest of people

- i18n (it is available to everyone, regardless of language)
- g11n (is available to everyone, regardless of culture)
- a11y (is available to everyone, regardless of ability)

Sustain???

At this point, it should be clear that "sustainability" is being overused and is somewhat semantically bleached.



When you see "Sustain", ask:

- For what?
- For whom?
- For how long?
- From what original state?
- Why this term?

Sustain covers all manner of sins.

- Was it healthy to begin with?
- Were people treated well?
- Were practices ethical?
- What about all of the stuff we talked about above?

What does "sustain" mean, anyway?

- Temporal term
- It ignores:
  - End-of-life cycles for software projects
  - End-of-life cycles for software ecosystems
  - Boom-and-bust cycles for fads



Garden Nasturtium https://www.inaturalist.org/observations/208389735

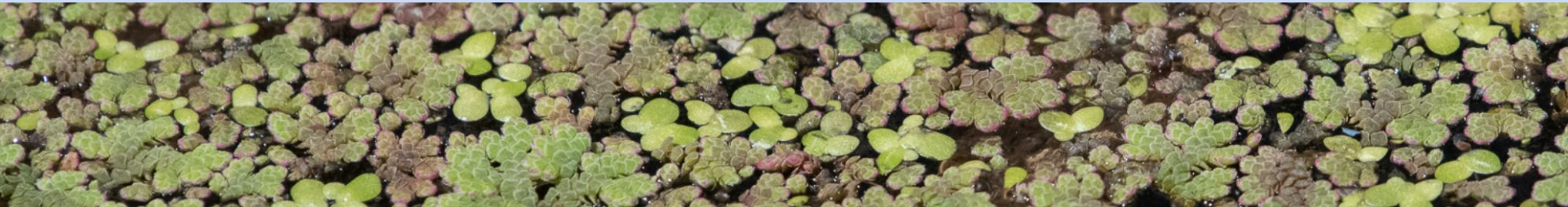
"Sustain" fails to recognize:

- Current achievements
- Creative output
- The bigger picture.

The bigger picture:

- Life on earth is dying.
- Things are mildly OK for some humans right now.
- The heat death of the universe.

Red Azolla https://www.inaturalist.org/observations/143949050



The bigger picture:

Every piece of software you love will die and be forgotten.

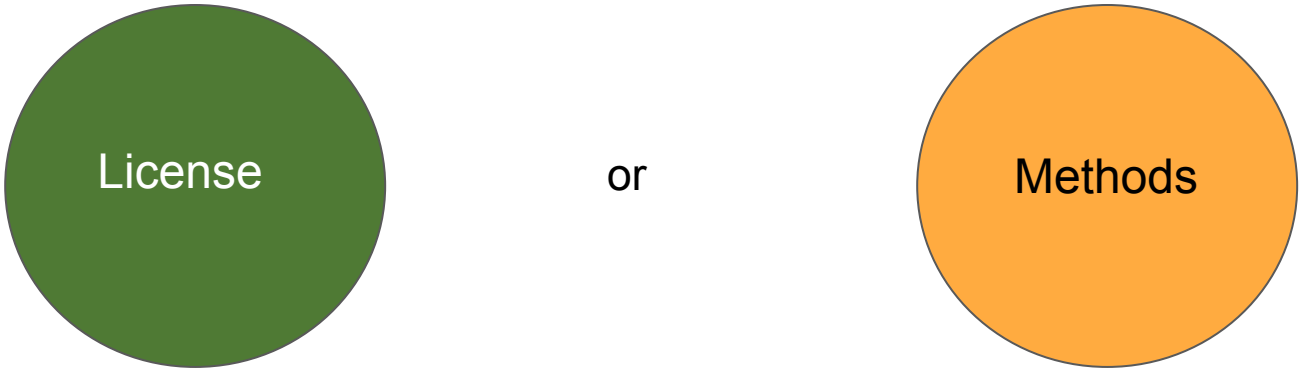
Baldcypress https://www.inaturalist.org/observations/146574140





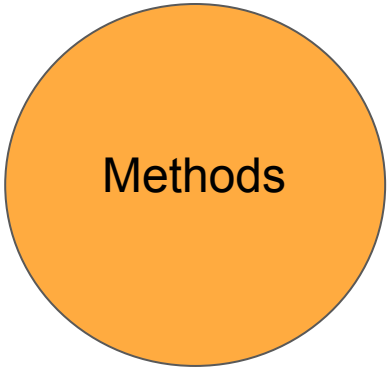
Source: Midjourney

Let's get back to open source:



Using this definition:

"Open source is a verb, not a noun."



So what is open, then?

- Antitrust maneuvering
- Open washing
- Avoiding sign-off from bureaucrats.
- Over-the-wall throwing
- AI scraping
- Extraction



Nuance

62 lines (37 loc) · 5.4 KB

Preview | Code | Blame

Raw

Contributing Code to iNaturalist

iNaturalist is open source for the sake of transparency, to take advantage of stuff that's free for open source projects, and so volunteer developers can help us out with small coding tasks if they'd like to. Unlike many open source projects, we do not intend it as a tool one can fork and customize for different applications. Instead, we code with only one instance in mind: iNaturalist.org. We mention this mainly to set expectations. We can't / won't stop you from forking this repo to make a celebrity spotting platform, but don't expect us to merge any of that code into this repo.

https://github.com/inaturalist/inaturalist/blob/main/CONTRIBUTING.md

Nuance

Open source? Sure!

62 lines (37 loc) · 5.4 KB

Preview | Code | Blame

Raw

Contributing Code to iNaturalist

iNaturalist is open source for the sake of transparency, to take advantage of stuff that's free for open source projects, and so volunteer developers can help us out with small coding tasks if they'd like to. Unlike many open source projects, we do not intend it as a tool one can fork and customize for different applications. Instead, we code with only one instance in mind: iNaturalist.org. We mention this mainly to set expectations. We can't / won't stop you from forking this repo to make a celebrity spotting platform, but don't expect us to merge any of that code into this repo.

https://github.com/inaturalist/inaturalist/blob/main/CONTRIBUTING.md

Conclusion

Not all open source communities are the same.

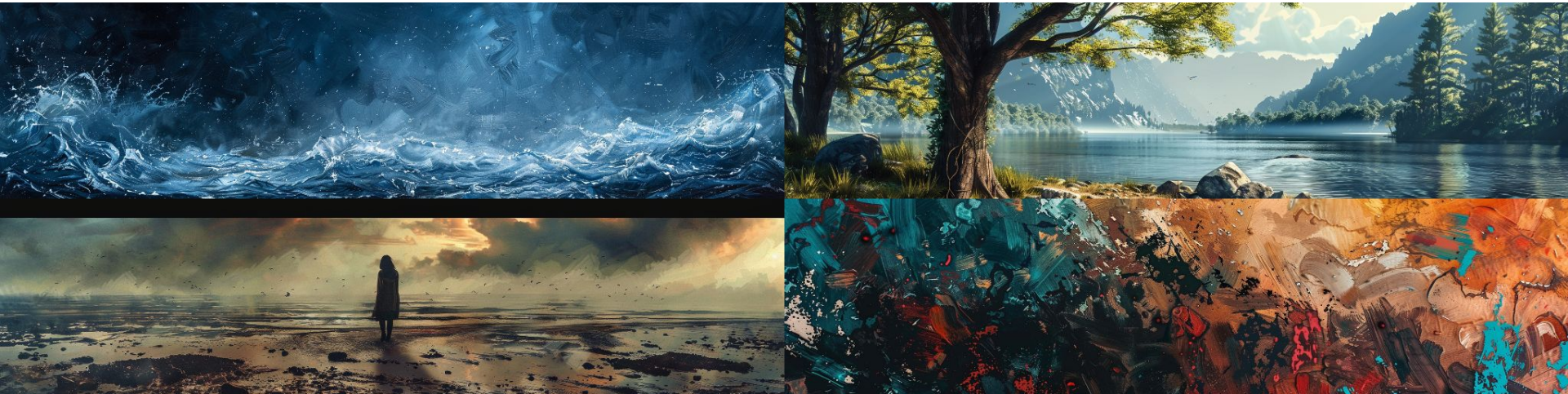
Conclusion

Not all open source is the same.

Conclusion

Not all open is the same.

"Open" from Midjourney



But…

Is this project OSI  approved: Yes or No.
Is it Open Source: Well…

Because this guy

- Ferdinand Saussure
- Inventor of semiotics

- Denotation
- Connotation



Source: Wikipedia

Because this guy

- Ferdinand Saussure
- Inventor of semiotics

- Denotation => OSI
- Connotation => Open source more broadly



Source: Wikipedia

Abstractions

- Depend upon shared context, values, and delegated authority
- Useful for talking to each each other



So, what matters?

- To act cooperatively and virtuously
- To make life more meaningful
- To reduce pain and suffering

Instead of saying "open source FTW", ask:

- Does my personal development life / project / ecosystem / business contribute to the happiness of myself and others?
- Does it recognize, validate, and protect basic living rights?
- Does it reduce harm?
- Does it increase capacity for wonder, awe, and joy?

What gets measured is what is valued.

- How are you measuring joy?
- How are you measuring wonder?
- How are you measuring impact on real lives?

How are you working towards improving those metrics?

Thanks.

Questions?



Unknown Fir https://www.inaturalist.org/observations/193983218